



Muntah Mahfuz Srestho

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Self-taught Machine Learning Engineer specializing in computer vision and NLP. Proven ability to build end-to-end ML pipelines, from training custom transformer models to deploying quantized object detection systems on edge devices. Adept at leveraging PyTorch and FastAPI to solve real-world backend and AI challenges in agile, remote environments.

TECHNICAL SKILLS

📄 Languages & Frameworks

Python, PyTorch, TensorFlow, FastAPI

📊 Data Science & ML

NumPy, Pandas, Matplotlib, Seaborn, HuggingFace

🔧 Tools & Deployment

Git, Docker, Render, Vercel, Netlify, Google Colab, Roboflow, Arch Linux

📐 Core Mathematics

Differential Calculus, Linear Algebra, Statistical Modeling

ENGINEERING PROJECTS

Custom Transformer & Language Models

Self-Directed

PyTorch • Python • NumPy • Statistical Modeling

🔗 github.com/Srestho0101/Trigram-Language-Model

- Engineered a custom Trigram Language Model in PyTorch from scratch, implementing tensor multiplication and statistical frequency analysis to achieve optimized validation loss.
- Developed a Bigram model implementing the Makemore architecture for efficient character-level text generation and predictive modeling.

Real-Time Object Detection Drone System

Active Development

YOLOv8n • Google Colab • Roboflow • Arch Linux

🔗 github.com/Srestho0101/monster-vision

- Designed an end-to-end edge-computing pipeline streaming live video to an Arch Linux server, achieving real-time inference using a dynamically quantized YOLOv8n model.

- Fine-tuned the YOLO architecture on a custom dataset, significantly optimizing inference speed and targeted detection accuracy for dynamic drone environments.

Backend API Pipeline for E-commerce

Active Development

Python • FastAPI • RAG Architecture

github.com/Srestho0101/data-retriever-bot

- Architecting a Retrieval-Augmented Generation (RAG) backend using FastAPI for a live E-commerce platform, enabling automated, real-time inventory querying.
- Developing tool-use logic to interface seamlessly with production databases, improving data retrieval efficiency and automated product updates.



EDUCATION & INDEPENDENT STUDY

Machine Learning Foundations

Self-Taught

Rigorous independent study in Differential Calculus, Linear Algebra, and Statistics specifically tailored for Machine Learning and AI applications.

Continuous Learning & Development

Self-Directed

Actively expanding expertise in modern backend deployments, containerization (Docker), and cloud ecosystems to complement ML engineering skills.